

Andres Lopez

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EDUCATION

Boston University

B.A. Computer Science | Minor in Business

Expected Graduation May 2028

GPA: 3.54

- Relevant Courses: Probability in Computing, Linear Algebra, Analysis of Algorithms, Computer Systems, Corporate Finance

SKILLS AND ACHIEVEMENTS

Languages: Python, Java, C, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, Next.js, Node.js, Express, Flask, FastAPI, Hono, Pandas, PyTorch, D3.js, Chart.js, OpenCV

AI & Agentic Development: Claude Code, Model Context Protocol (MCP), LLM APIs (Anthropic, Gemini, OpenAI), RAG, agentic evals

Tools & Infrastructure: Git, GitHub Actions, Docker, Cloudflare (Workers, D1, KV, R2), PostgreSQL, Supabase, Stripe Connect, RESTful APIs

Achievements: 4x Hackathon Wins, Eagle Scout, Google Software Engineer Program (G-SWEP), MLT CP Fellow, HSF Scholar

EXPERIENCE

Software Engineer Intern

Jul 2026 – Present

Grove Tax (a16z Speedrun)

Hybrid

- Re-architected the platform from single-purpose tax-return software into a multi-engagement system supporting 12+ engagement types including bookkeeping, extensions, and extended returns, shipping 3 tax-return delivery flows on the new model as 1 of 5 engineers
- Engineered Stripe Connect payments infrastructure for client tax preparation, implementing connected-account onboarding, webhook handling, and payout flows across the end-to-end payment path, currently in staged rollout across 3 design-partner firms
- Built AI-assisted document extraction tooling with Google Gemini on AWS spanning 12+ taxpayer document types, gating extracted values behind reviewer verification before they reached downstream return calculations

AI/ML Fellow

Mar 2026 – Present

Break Through Tech | Cornell Tech

Remote

- Completed a 9-week ML Foundations course toward a machine learning certificate from Cornell University, covering preprocessing, feature engineering, and model selection; team-based industry challenge project underway

Software Engineer

Dec 2025 – Present

Hack4Impact at Boston University

Boston, Massachusetts

- Engineered 15+ Flask RESTful API endpoints and architected 5+ Supabase schemas with relational integrity for 2 nonprofits, directly supporting 250+ community members
- Led backend development across 10+ features on a 6-person agile team, owning API architecture, database optimization, and 230+ integration tests, and delivering every feature on schedule

Data Analytics Intern

Jun 2025 – Aug 2025

Kearny Bank

Fairfield, New Jersey

- Analyzed 5+ years of historical banking data using Python and SQL to optimize 8 ETL pipelines, reducing organization-wide query execution times by 15% through bottleneck identification and query optimization
- Built 5 Power BI dashboards used weekly by 20+ stakeholders to track ETL performance metrics, implementing automated Python scripts for database monitoring and validation across multiple banking databases

PROJECTS

Sapling | CivicHacks BU '26, AI Tutor Track | Python, Next.js, FastAPI, Supabase, Google Gemini

Mar 2026 – Present

- Led backend development on a 4-person AI study platform spanning 197 API endpoints, a 56-table Postgres schema, 18 LLM agents, and 2,000+ backend tests across 57 append-only migrations
- Built an end-to-end testing harness in which an LLM explores the running app as a user while 6 deterministic Python oracles, never the model, adjudicate correctness, surfacing 7 bugs mocked tests could not detect, plus AES-256-GCM encryption across 34 columns with numeric-preserving reads

Canopy | Cloudflare Workers, Hono, D1, Model Context Protocol

Jun 2026 – Present

- Sole-authored a documentation and context store for AI coding agents: one Cloudflare Worker serving a Hono REST API, a stateless MCP endpoint, and a TypeScript SPA over D1, with FTS5 BM25 retrieval and capability-scoped tools, distributed to the team as a Claude Code plugin bundling three agent skills (15,500 LOC, 439 tests)
- Designed an idempotent write gate for untrusted agent-generated writes using a replay ledger keyed on session and item index, SHA-256 content hashing, and LCS diff edit/rewrite classification, with confirm actions exposed only as authenticated HTTP routes so agents can only stage

Recost | Stripe Best Web API, HackIllinois '26 | TypeScript, Cloudflare Workers, Hono, D1, React

Feb 2026 – Jun 2026

- Co-founded and incorporated an API cost-intelligence platform (Delaware C-corp) with a 3-person team, architecting a serverless REST API with 30+ endpoints on Cloudflare Workers (Hono, D1) with Google OAuth, API key management, and KV-based rate limiting
- Built a VS Code extension using web-tree-sitter to detect N+1 hotspots, batching opportunities, and cross-file call origins across JS/TS/Python, plus Node.js and Python SDKs intercepting outbound HTTP at runtime to capture cost and latency telemetry across 8+ providers